

Team HomeSick

RoomLagbe

Where

Select Radius ▾

Uni Location

Choose Location

Price Range

Min – Max ▾

Search

CSE- 4508
RDBMS PROJECT
05.03.2026



OUR TEAM

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BRIEF OVERVIEW

RoomLagbe is a student focused housing platform designed to help university students find affordable rentals and trustworthy roommates near campus. The system supports:



**Apartment
listings by owners**



**Smart search with
filters and commute
time estimation**

**Student-to-student
room sharing**

**Secure visit request
and approval
workflow**

**Role-based access
for students and
owners**



FEATURES

Authentication & Roles

- Student, Owner, Admin roles
- Secure login/signup
- Profile management
- Role-based access control & dashboards

Verification System

- Student & Owner document verification
- Admin approval/rejection panel
- Blue verified badge for lister
- Verification notifications

Listing Management

- Create apartment/room-share listings
- Image upload with preview
- Full listing detail view
- Per listing stats (owner side)

Smart Search & Map Discovery

- Advanced filtering
- OpenStreetMap location autocomplete
- Radius-based nearby search
- Interactive map with markers



FEATURES

Commute Calculator

- University/destination search
- Road-based distance & time estimation
- Walking & driving time
- Mini route map

Rent Analytics

- Fair-rent score (1–5)
- Area-based rent comparison
- Radius-wise rent distribution
- Admin analytics dashboard

Visit Booking System

- Visit request & approval
- 2-hour time blocking
- Booking status tracking

Notification System

- Visit updates
- Verification approval alerts
- Bell icon with unread count

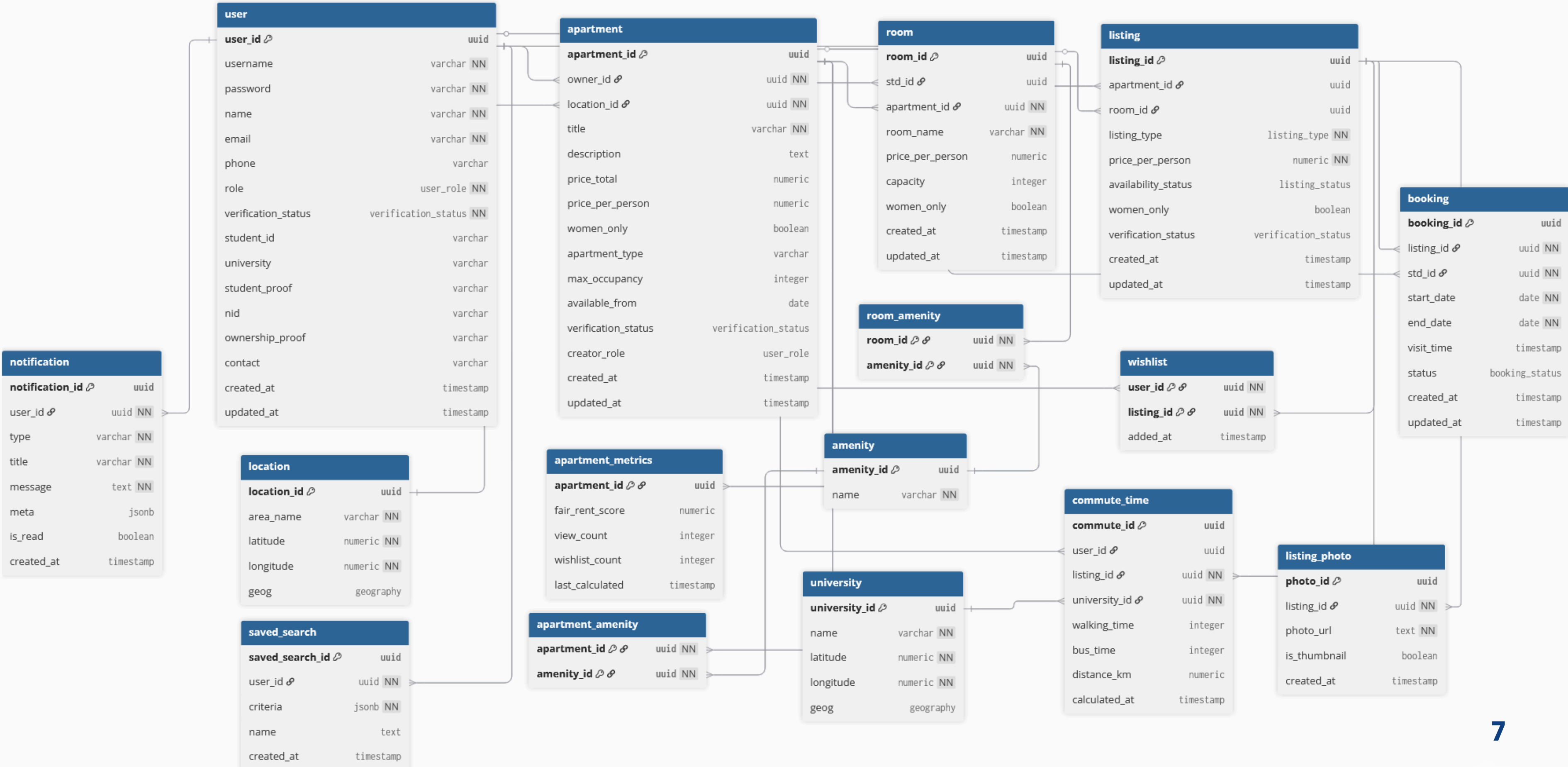


TOOLS & TECHNOLOGY

- **Frontend:** React.js
- **Backend:** Node.js + Express.js
- **Database:** PostgreSQL + PostGIS
(Using Supabase)
- **Authentication:** JWT
- **Styling:** Native CSS
- **Mapping / Commute:** OpenStreetMap API
- **Testing:** Postman
- **Version Control:** Git + GitHub



DATABASE DESIGN





IMPLEMENTED TRIGGERS

Trigger Name	Timing	Table	Purpose
trg_user_update_at	Before Update	User	Automatically updates timestamp whenever a record is modified
trg_location_set_geog	Before Insert or Update	Location	Automatically converts latitude/longitude coordinates into PostGIS geography points.
trg_university_set_geog	Before Insert or Update	University	Converts university coordinates into PostGIS geography points for spatial calculations.
trg_apartment_updated_at	Before Update	Apartment	Tracks when apartment details are modified (price, description, amenities, etc.)



IMPLEMENTED TRIGGERS

Trigger Name	Timing	Table	Purpose
trg_room_updated_at	Before Update	Room	Tracks when room details are modified (price, capacity, etc.)
trg_listing_updated_at	Before Update	Listing	Tracks when listings are modified (status, price, availability)
trg_booking_updated_at	Before Update	Booking	Tracks when booking status changes (pending → approved/rejected)



IMPLEMENTED FUNCTIONS

Type : Trigger

**update_modified_
column**

- Purpose: Auto-updates `updated_at`
- Used in: user, apartment, room, listing, booking

location_set_geog

- Purpose: Convert latitude/longitude → PostGIS geography
- Used in: location

**university_set_
geog**

- Purpose: Convert university coordinates → geography
- Used in: university



IMPLEMENTED FUNCTIONS

Type : Query

get_listings_near_location

- **Type:** Set-Returning Function
- **Returns:** TABLE(listing_id, distance_meters)
- **Purpose:** Radius-based listing search
- **Uses** PostGIS spatial queries

calculate_listing_university_distance

- **Type:** Scalar Function
- **Returns:** Distance in kilometers
- **Purpose:** Show distance between listing & university



POSTGIS SPATIAL FEATURES

PostGIS is an extension for PostgreSQL that adds spatial (geographic) capabilities to the database.

- **Geography columns on location and university tables**
- **Automatic coordinate conversion**
- **Spatial indexes (GIST) for performance**
- **Distance calculations in meters/kilometers**
- **Radius-based search support**



INDEXES & CONSTRAINTS

B-Tree Indexes

- idx_user_username | idx_user_email | idx_user_role
- idx_listing_type | idx_listing_status | idx_listing_price | idx_listing_women_only
- idx_booking_listing_id | idx_booking_user_id
- idx_apartment_owner_id | idx_room_apartment_id
- idx_listing_apartment_id | idx_listing_room_id
- idx_apartment_location_id | idx_commute_listing_id | idx_commute_university_id

Unique Indexes / Constraints

- uq_user_username | uq_user_email | uq_user_phone
- uq_amenity_name | uq_listing_thumbnail | uq_commute_listing_university

Spatial (GIST) Indexes

- idx_location_geog
- idx_university_geog



GITHUB LINK & COMMITS

Prince02031/ RoomLagbe



RoomLagbe is a student-focused apartment rental platform that enables landlord listings and student-to-student room sharing, with fair rent estimation, walking-time...

3 Contributors 0 Issues 0 Stars 0 Forks

Prince02031/RoomLagbe: RoomLagbe is a student-focused apartment rental platform that enables landlord listings and student-to-student room...

RoomLagbe is a student-focused apartment rental platform that enables landlord listings and student-to-student room sharing, with fair rent estimation, walking-time commute calculation, and secure ...



github.com/Prince02031/RoomLagbe

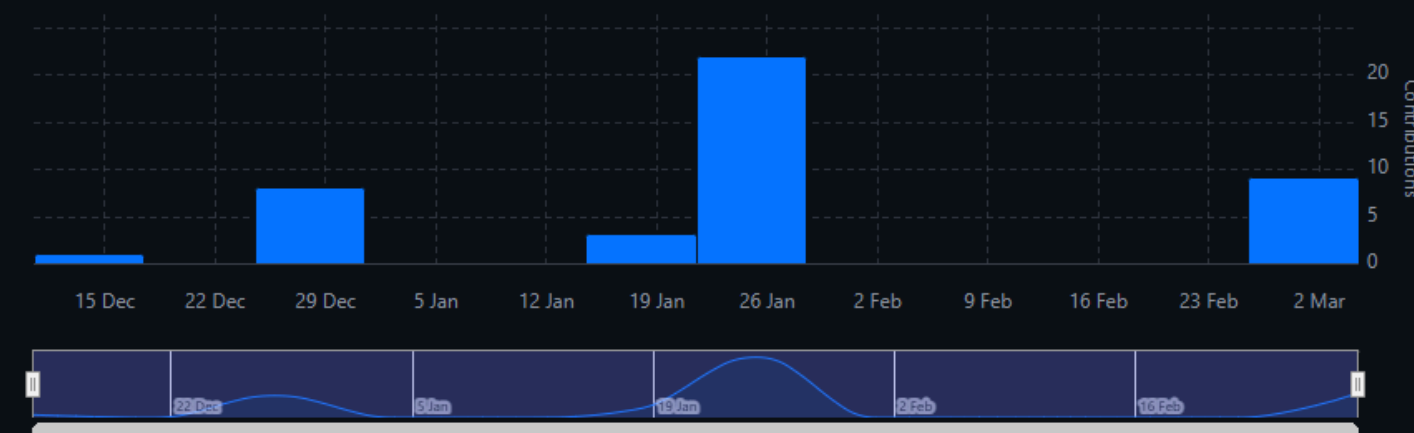
Contributors

Contributions per week to main, excluding merge commits

Period: All Contributions: Commits

Commits over time

Weekly from Dec 14, 2025 to Mar 1, 2026



alfishahrin-05
22 commits 11,679 ++ 1,117 --



Rahatut
11 commits 15,644 ++ 8,271 --



Prince02031
10 commits 403,593 ++ 393,268 --





DEMONSTRATION



INDIVIDUAL CONTRIBUTIONS

Member	Database	Backend	Frontend
Rahatut Tahrim Mounota	Initial Schema design, ER diagram, normalization, indexing, alter table user	Authentication, Initial Models, Controllers & Routes for core features	Listing creation and display pages (Homepage, apartment/room cards etc), Admin workflow & dashboard, verification
Nayeemul Hasan Prince	Final Schema design, Triggers, Functions, data integrity & validation rules, added notif table	Updated Models, Core Business Logic (services) and Backend DB Integration, notification model, controller, routes	Owner Dashboard, Listing Details page, Amenities Selection & Location Dropdown , map, fair rent, notifications
Alfi Shahrin	Functions, Audit Log Design, Views, analytics queries	Bookings & Wishlists, Protected Routes & Postman Collections, multer, photo upload endpoint	Student Dashboard, Booking Interfaces, saved searches, Profile page, image upload, commute time using polyline



THANK YOU